1. Lesson 7: Harmonic osicllator and Hermite polynomials

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As a subproblem of the boundary value problem solved by separation of variables, there is an *eigenvalue problem* of the following type:

$$\begin{cases} y''(x) + \lambda y(x) = 0, & x \in [0, L] \\ y(0) = 0 = y(L) \end{cases} \tag{1.1}$$

where, a priori, both the function $y(x)$ and the number λ are unknown. It is found that necessarily it must be

$$\lambda_n = \frac{n^2\pi^2}{L^2}, \qquad n = 1, 2, 3, ...$$
$$y_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

The eigenvalue problem therefore has:

1) A sequence of positive eigenvalues $\{\lambda_n\}_{n=1}^{\infty}$ that tends to $+\infty$.
2) A corresponding sequence of y_n, which forms an orthogonal system in $L^2[0, L]$.

There exists a general theory of eigenvalue problems for ordinary differential equations, of which the example just discussed is a particular case. This theory leads to the construction of complete orthogonal systems in $L^2[a, b]$ consisting of eigenfunctions. We provide a brief overview of this theory.

Definition 1.1 (Strum-Liouville problem)**.** Consider the eigenvalue problem, known as the Sturm-Liouville problem:

$$\begin{cases} \frac{d}{dx}\left(p(x)\frac{du}{dx}\right) - q(x)u + \lambda\rho(x)u = 0, & x \in [a, b] \\ u(a) = u(b) = 0 \end{cases} \tag{1.2}$$

where $a, b \in \mathbb{R}$ and the functions p, q, and ρ are given coefficients satisfying the following assumptions:

- $p \in C^1[a, b]$, $p(x) > 0$
- $q \in C^0[a, b]$, $q(x) \geq 0$
- $\rho \in C^0[a, b]$, $\rho(x) > 0$

Under the above hypothesis the Strum-Liouville problem is called regular.

Remark 1.2. The problem (1.1) is a classical regular Strum-Louville problem with $p = \rho = 1$, $q = 0$.

The following result is very important in the regular Strum-Liouville problems.

Theorem 1.3. *Given a regular Sturm-Liouville problem* (1.2)*, there exist:*

1) A sequence of eigenvalues $\{\lambda_n\}_{n=1}^{\infty}$, all positive, increasing, and tending to $+\infty$;
2) A corresponding sequence of eigenfunctions $\{u_n\}_{n=1}^{\infty}$, which form a complete orthonormal system in the Hilbert space $L^2\left((a, b), \rho(x)dx\right)$.

Explicitly, this means that the measure is the weighted measure $d\mu = \rho(x)dx$, so orthogonality means that

$$\int_a^b u_n(x)u_m(x)\rho(x)dx = \begin{cases} 0, & n \neq m \\ 1, & n = m \end{cases}$$

However, the Sturm-Liouville problems that arise from applications often do not satisfy all the assumptions of the previous theorem (and are called *singular Sturm-Liouville problems*)

1.1. **Hermite equation.** One of the singular Sturm-Liouville problems, which we will use to study the quantum harmonic oscillator, is the Hermite equation. This equation is given by

$$y'' - 2xy' + \lambda y = 0, \qquad x \in \mathbb{R}, \tag{1.3}$$

which can be written in the following form

$$(e^{-x^2} y')' + \lambda e^{-x^2} y = 0.$$

This is a Sturm-Liouville problem on $\mathbb{R}$ with $p(x) = \rho(x) = e^{-x^2}$, $q(x) = 0$ made singular by the fact that the equation is not defined on a segment but on the entire line.

Theorem 1.4. *The eigenvalues of* (1.3) *are non-negative; eigenfunctions corresponding to distinct eigenvalues are orthogonal in* $L^2\left(\mathbb{R}, e^{-x^2} dx\right)$.

Theorem 1.5. *The Hermite equation has solutions in* $L^2\left(\mathbb{R}, e^{-x^2} dx\right)$ *only for a sequence of eigenvalues* $\lambda_n = 2n$ *for* $n = 0, 1, 2, 3, \ldots$. *The solution* $H_n(x)$ *of the Hermite equation, corresponding to* $\lambda_n = 2n$ *and normalized in* $L^2\left(\mathbb{R}, e^{-x^2} dx\right)$, *is a polynomial of degree* n. *The system* $\{H_n(x)\}_{n=0}^{\infty}$ *forms a complete orthonormal basis in* $L^2\left(\mathbb{R}, e^{-x^2} dx\right)$.

Explicitly, the theorem states that

$$\left(e^{-x^2} H_n(x)\right)' + 2ne^{-x^2} H_n(x) = 0$$

with

$$\int_{\mathbb{R}} H_n(x) H_m(x) e^{-x^2} dx = \begin{cases} 1, & m = n \\ 0, & m \neq n. \end{cases} \tag{1.4}$$

It is possible to write the first Hermite polynomials by orthonormalizing the functions 1, x, x^2, x^3,... in the space $L^2\left(\mathbb{R}, e^{-x^2} dx\right)$. We find

$$H_0(x) = \frac{1}{\pi^{1/4}}$$

$$H_1(x) = \frac{\sqrt{2}}{\pi^{1/4}} x,$$

$$H_2(x) = \frac{-2 + 4x^2}{2\sqrt{2}\pi^{1/4}},$$

$$\ldots$$

The following explicit formula can be proved:

$$H_n(x) = \frac{(-1)^n}{\sqrt{\sqrt{\pi} 2^n n!}} e^{x^2} \frac{d^n}{dx^n} \left(e^{-x^2}\right).$$

Thus, given $f \in L^2\left(\mathbb{R}, e^{-x^2} dx\right)$, its Fourier coefficients with respect to the system $\{H_n(x)\}_{n=0}^{\infty}$ are given by

$$c_n = \int_{\mathbb{R}} f(x) H_n(x) e^{-x^2} dx \tag{1.5}$$

and lead to the expansion

$$f(x) = \sum_{n=0}^{\infty} c_n H_n(x). \tag{1.6}$$

1.2. **Quantum harmonic oscillator.** In quantum mechanics, a particle of mass m moving along a line is described by a state function $\psi(x,t)$. Given this function, the probability that the particle is found at time t within the interval (a,b) is given by the integral

$$\int_a^b |\psi(x,t)|^2 dx.$$

Due to its probabilistic interpretation, $\psi(x,t)$ must satisfy the normalization condition

$$\int_{\mathbb{R}} |\psi(x,t)|^2 dx = 1, \qquad \forall t \in \mathbb{R}. \tag{1.7}$$

The function $\psi(x,t)$ satisfies the Schrödinger equation, which, in the case of a particle of mass m subject to a potential field $V(x)$, is:

$$i\hbar \frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2}\psi(x,t) + V(x)\psi(x,t), \quad x \in \mathbb{R},\ t \in \mathbb{R}, \tag{1.8}$$

where $\hbar := \frac{h}{2\pi}$ and h is the Planck constant. We look for solutions using the method of separation of variables:

$$\psi(x,t) = X(x)T(t).$$

We plug this into the equation (1.8) and we get

$$i\hbar XT' = -\frac{\hbar^2}{2m}X''T + V(x)XT$$

We divide by $X(x)T(t)$ and we obtain

$$i\hbar\frac{T'}{T} = -\frac{\hbar^2}{2m}\frac{X''}{X} + V(x).$$

From which we get the following ODEs

$$i\hbar\frac{T'}{T} = E, \tag{1.9}$$

$$-\frac{\hbar^2}{2m}\frac{X''}{X} + V(x) = E, \tag{1.10}$$

where $E \in \mathbb{R}$ is a constant. The solution of the ODE (1.9) is given by

$$T(t) = ce^{-\frac{iE}{\hbar}t}, \tag{1.11}$$

where c is a constant. Since $|T(t)| = c$, the condition on the normalization condition, given by (1.7), becomes

$$1 = \int_{\mathbb{R}} |\psi(x,t)|^2 dx = |T(t)|^2 \int_{\mathbb{R}} |X(x)|^2 dx = c^2 \int_{\mathbb{R}} |X(x)|^2 dx \quad \Rightarrow \int_{\mathbb{R}} |X(x)|^2 dx = \text{const.} \tag{1.12}$$

We are therefore interested in determining the eigenfunctions in $L^2(\mathbb{R})$ of the equation (1.10) that can be rewritten as

$$X'' + \frac{2m}{\hbar^2}\left(E - V(x)\right)X = 0, \quad \text{in} \quad \mathbb{R}. \tag{1.13}$$

Let us now consider the particular case of the *quantum harmonic oscillator* (one-dimensional), given by

$$V(x) = \frac{1}{2}m\omega^2 x^2.$$

The quantum harmonic oscillator model describes, as a first approximation, the oscillations of a system near a stable equilibrium point.
The equation (1.13) can be written as

$$X''(x) + \left(\frac{2m}{\hbar^2}E - \frac{m^2\omega^2x^2}{\hbar^2}\right)X(x) = 0. \tag{1.14}$$

To solve it, the idea is to reduce it to the Hermite equation, which we encountered in the first part of the class. The procedure consists of two steps.

1) First, we want to perform a change of variable that makes the coefficient of x^2 equal to 1. Let $X(x) = Z(\alpha x)$, with α to be determined, then

$$\alpha^2 Z''(\alpha x) + \left(\frac{2m}{\hbar^2}E - \frac{m^2\omega^2x^2}{\hbar^2}\right)Z(\alpha x) = 0.$$

We set $\alpha x = y$ and we have

$$\alpha^2 Z''(y) + \left(\frac{2m}{\hbar^2}E - \frac{m^2\omega^2}{\alpha^2\hbar^2}y^2\right)Z(y) = 0.$$

Now, we want

$$\alpha^2 = \frac{m^2\omega^2}{\alpha^2\hbar^2} \Rightarrow \alpha^4 = \frac{m^2\omega^2}{\hbar^2} \Rightarrow \alpha = \sqrt{\frac{m\omega}{\hbar}}.$$

Hence we have that

$$X(x) = Z\left(\sqrt{\frac{m\omega}{\hbar}}x\right), \tag{1.15}$$

solves the equation (1.14) if and only if $Z(y)$ solves the equation:

$$\frac{m\omega}{\hbar}Z''(y) + \left(\frac{2m}{\hbar^2}E - \frac{m\omega}{\hbar}y^2\right)Z(y) = 0.$$

We multiply by $\frac{\hbar}{m\omega}$ and we get

$$Z''(y) + \left(\frac{2}{m\hbar}E - \ y^2\right)Z(y) = 0. \tag{1.16}$$

2) Now, we transform this equation into the Hermite equation by setting:

$$Z(y) = e^{-\frac{y^2}{2}}Y(y).$$

The first derivative is given by

$$Z'(y) = e^{-\frac{y^2}{2}}\left(-yY(y) + Y'(y)\right).$$

The second derivative can be written as

$$Z''(y) = e^{-\frac{y^2}{2}}\left(y^2Y(y) - yY'(y) - Y(y) - yY'(y) + Y''(y)\right) = e^{-\frac{y^2}{2}}\left(y^2Y(y) - 2yY'(y) - Y(y) + Y''(y)\right).$$

Thus by (1.16) we have

$$e^{-\frac{y^2}{2}}\left(y^2Y(y) - 2yY(y) - Y(y) + Y''(y) + \left(\frac{2}{m\hbar}E - y^2\right)Y(y)\right) = 0.$$

Hence we have

$$Y''(y) - 2yY'(y) + \left(\frac{2}{\omega\hbar}E - 1\right)Y(y) = 0. \tag{1.17}$$

Hence $Z(y)$ solves (1.16) if and only if $Y(y)$ solves (1.17). The normalization condition on Y is read as follows:

$$\begin{aligned}\infty > \int_{\mathbb{R}} |X(x)|^2 dx &= \int_{\mathbb{R}} \left| Z\left(\sqrt{\frac{m\omega}{\hbar}}x\right)\right|^2 dx \\ &= \sqrt{\frac{\hbar}{m\omega}} \int_{\mathbb{R}} |Z(y)|^2 dy \\ &= \sqrt{\frac{\hbar}{m\omega}} \int_{\mathbb{R}} e^{-y^2} |Y(y)|^2 dy.\end{aligned}$$

Therefore, we are interested in finding eigenfunctions Y and eigenvalues λ of the Hermite equation.

$$Y'' - 2yY' + \lambda Y = 0, \qquad \lambda := \frac{2}{\omega h}E - 1$$

for which we have

$$\int_{\mathbb{R}} e^{-y^2} |Y(y)|^2 dy < \infty \Rightarrow Y(y) \in L^2\left(\mathbb{R}, e^{-y^2}\right).$$

From Theorem 1.5, it follows that this condition can only be satisfied for $\lambda = 2n$ and $Y(y) = H_n(y)$, the n-th Hermite polynomial. This means that

$$\frac{2}{\omega\hbar}E - 1 = 2n,$$

By relabelling E with E_n we get

$$E_n = \left(n + \frac{1}{2}\right)\omega\hbar, \qquad n = 0, 1, 2, 3, ... \tag{1.18}$$

and the corresponding eigenfunctions are

$$Y_n(y) = H_n(y) \Rightarrow Z_n(y) = e^{-\frac{y^2}{2}} H_n(y),$$

and therefore for the initial equation we have (see (1.15)):

$$X_n(x) = e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right). \tag{1.19}$$

Now, we compute the $L^2(\mathbb{R}, e^{-y^2})$-norm of the Hermite polynomials $H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right)$. By (1.4) we have

$$\begin{aligned}\|H_n\|^2_{L^2(\mathbb{R},e^{-y^2})} &= \int_{\mathbb{R}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}y\right) H_n\left(\sqrt{\frac{m\omega}{\hbar}}y\right) e^{-\frac{m\omega}{\hbar}y^2} dy \\ &= \sqrt{\frac{\hbar}{m\omega}} \int_{\mathbb{R}} H_n(x) H_n(x) e^{-x^2} \\ &= \sqrt{\frac{\hbar}{m\omega}}.\end{aligned}$$

If the Hermite polynomials H_n are normalized to have unit norm in $L^2(\mathbb{R}, e^{-y^2}dy)$, then the normalized eigenfunction will be [1]

$$X_n(x) = \sqrt[4]{\frac{m\omega}{\hbar}} e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right), \qquad n = 0, 1, 2, 3, ...$$

[1] We are just dividing X_n (see (1.19)) by $\|H_n\|_{L^2(\mathbb{R},e^{-y^2})}$

Now we substitute (1.18) in (1.11) and we have

$$T(t) = c_n e^{-i\left(n+\frac{1}{2}\right)\omega t}.$$

The equation in (x,t) therefore has solutions in separated variables.

$$\psi_n(x,t) = c_n e^{-i\left(n+\frac{1}{2}\right)\omega t} \sqrt[4]{\frac{m\omega}{\hbar}} e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right)$$

By the superposition principle the solution of the Schrödinger equation with initial condition $\psi(x,0)$ is then given by the series

$$\psi(x,t) = \sum_{n=0}^{\infty} c_n e^{-i\left(n+\frac{1}{2}\right)\omega t} \sqrt[4]{\frac{m\omega}{\hbar}} e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right),$$

with

$$\psi(x,0) = \sum_{n=0}^{\infty} c_n \sqrt[4]{\frac{m\omega}{\hbar}} e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right),$$

where by (1.5) and (1.6) the coefficients are given by

$$c_n = \int_{\mathbb{R}} \psi(z,0) \sqrt[4]{\frac{m\omega}{\hbar}} e^{-\frac{m\omega z^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} z\right) dz.$$